LiDAR-Recovered Body Pose Does Not Improve Pedestrian Trajectory Forecasting on Waymo, and the Tracker Already Explains Why

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***Abstract*—Body pose is an intuitively attractive input for pedestrian trajectory forecasting: gait, stance and facing appear to encode intent that a position history does not. We test this on the Waymo Open Motion Dataset with SMPL poses recovered from LiDAR, and find no measurable benefit. The decisive comparison is not pose against no pose—which confounds body information with the capacity of an extra input branch—but correct pose against the same architecture fed *another pedestrian's* pose. On that control the effect is $+0.66\%$ (p=0.52) from scratch, $+0.88\%$ (p=0.18) after full-dataset pretraining, -1.02% (p=0.56) at an intermediate scale and $+0.04\%$ (p=0.93) at $43\times$ more training scenes, while the baseline improves 19% from data alone; at the largest scale the data bound the content effect to $\pm 1.7\%$. A directed search over the integration code accepted one of 22 candidates under a truncated proxy, and the pre-registered eight-seed confirmation that followed returned -0.04% (p=0.92). An oracle control rules out a dead pathway: the same branch, fed the agent's own future instead of its body, improves minADE by 34.8%. We then explain the null rather than report it. Body facing does predict which way a standing pedestrian steps off, at roughly twice chance—but the tracker's box heading estimates the same quantity more accurately (25.2° versus 44.2° median error), is already an input to the model, and is not beaten by pose in any range, occlusion or heading-jitter stratum; the two degrade together because both derive from the same returns. Head orientation, the one potentially complementary cue, is under-dispersed fivefold by the fitted meshes (3.8° against a 19.2° ground-truth spread). Finally, we show that aggregate MPJPE is the wrong acceptance metric for pose in a perception stack: scored per joint against a mean-pose baseline, the head carries a tenth of the skill of any limb joint while its MPJPE looks *better* than either wrist.**

***Index Terms*—pedestrian trajectory prediction, human pose estimation, LiDAR, autonomous driving, negative results**

I. Introduction

Predicting where a pedestrian will go is a safety-critical function of an automated vehicle, and body pose is an obvious candidate input. A person's stance, gait phase and facing direction plausibly reveal intent before their position changes, and modern perception stacks already produce human mesh or keypoint estimates. Several recent systems condition trajectory forecasts on pose.

This paper reports that, on the Waymo Open Motion Dataset (WOMD), pose conditioning does not measurably improve pedestrian trajectory accuracy—and, more usefully for practitioners, identifies the specific reason, which is a property of the perception stack rather than of the model.

We make three contributions.

A controlled null. We measure pose conditioning against a shuffled-pose control: the same architecture receiving a different pedestrian's body motion. This isolates information content from the capacity cost of an extra branch, and it is the comparison a practitioner should demand before adopting any auxiliary modality. The control is null at three training scales, across six pose encodings and two integration styles, under a directed architecture search, and in a pre-registered eight-seed confirmation on a held-out split. An oracle arm, which replaces the body with a signal encoding the agent's own future, improves minADE by 34.8% through the same branch—so the null is a property of pose, not of the plumbing.

A detection floor. We then measure how accurate a side channel must be before this pathway converts it into accuracy at all, by corrupting that oracle with a controlled angular error. The improvement halves for every 4.4° of bearing error and is gone by $\approx 15^\circ$, past which the branch is a net cost. SMPL body facing is accurate to 44.2° on the cue it is supposed to carry—about eight halvings below the useful range. The null is not a near miss, and the floor can be measured before any modality is integrated.

A measurement caveat. Aggregate MPJPE cannot distinguish a pose estimator that tracks a joint imprecisely from one that ignores the input and emits its prior. Scoring each joint against an always-predict-the-mean-pose baseline, MPJPE rank-orders joints roughly backwards with respect to information content. Accepting a pose estimator into an AV stack on a single error number risks importing a prior.

We also report, as a cautionary note, that our first attempt at this study was invalidated by an export bug that silently substituted other pedestrians' bodies—a failure any pose pipeline keyed by object identity can reproduce.

II. Related Work

Trajectory forecasting on WOMD is dominated by anchor- and query-based transformers; we build on MTR [1], [2],

which is representative and strong. Human mesh recovery from LiDAR is maturing [3], with motion priors used to complete sparse observations [4]–[6]. Intent-focused work on pedestrian datasets [7], [8] targets crossing decisions rather than metric trajectories.

Relation to prior positive results. Pose conditioning has reported real gains, and our mechanism predicts where they come from rather than contradicting them. Social-Transmotion [9] lowers ADE on the synthetic JTA dataset from 0.99 to 0.89 with *ground-truth* 3D pose, and—the harder case for us—from 0.40 to 0.36 on JRDB with *estimated* pose. Social-Pose [10] and Salzmann et al. [11] report improvements of the same order, the latter specifically for agents with little position history, which is also our best hypothesis. Three properties separate those settings from an AV stack, and all three follow from the mechanism in Sec. V. *Range and sensor:* JRDB is an indoor and campus robot dataset in which pedestrians are metres away and pose is camera-derived, whereas WOMD pedestrians sit at 20–75 m and return a few dozen LiDAR points. *The baseline:* those predictors consume positions, not a tracker's box heading, so body facing is new information there and duplicated information here—this, not the dataset, is what our stratified measurements implicate. *The control:* none of [9]–[11] reports a shuffled-pose arm, so a 4–10% gain is not separated from the capacity of the branch carrying it, and the branch tax we measure ($+1.7\%$ minADE, $+1.0\%$ minFDE) is of that same order. Our contribution is not an architecture but a controlled measurement of whether the modality pays in an AV setting, with the reason it does not.

III. Setup

A. Data and a data-integrity failure

We use WOMD [12] with SMPL [13] poses recovered from LiDAR by the Waymo-3DSkelMo pipeline [14], which fits per-frame bodies with LiDAR-HMR [3] and completes them using AMASS-trained motion priors [4]–[6]. Poses cover the 11 observed frames; the prediction horizon is 8 s at 10 Hz. The branch is not starved: every predicted agent carries at least one valid pose frame, and 86.7% of observed frames carry one, so sparsity of pose is not an explanation for anything that follows.

One property of that pipeline bears on Sec. V: when a pedestrian is moving and the raw LiDAR estimate of root rotation deviates from the past-trajectory direction by more than $90°$, the pipeline snaps the root to the direction of travel. Body facing is therefore partly trajectory-derived for moving pedestrians. This reinforces the redundancy we report rather than weakening it, and it does not reach the stationary-onset analysis, whose events are all below 0.3 m/s.

Our first pass at this study was invalid. The export keyed pose clips by a scene-*local* object identifier across $\sim$326k scenes, so clips collided across scenes: nearly every training sequence carried another pedestrian's body motion, and only about one pedestrian in five survived the collision ($4.7\times$ fewer than the corrected export). A per-scene re-export (299,220 scenes, 1.81M sequences) cut the median error between root heading and travel direction from $41.5°$ to $11.4°$ and raised the gait–speed correlation from $r{=}0.26$ to 0.58. We report this because the failure is silent—training converges, losses look normal—and because identity-keyed joins are common in pose pipelines. Every number in this paper is from the corrected export.

B. Arms and the content test

Three arms share one architecture (Fig. 1). *No-pose* receives trajectories and map. *Pose* adds the pedestrian's own SMPL sequence. *Shuffled* receives another pedestrian's sequence, with identical parameter count, optimiser and input dimensionality. A fourth arm, *oracle*, replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s, leaving the branch, its parameter count, its input dimensionality and its validity masks untouched; it exists to show that the pathway can carry information at all (Sec. IV-D).

Pose versus no-pose confounds information with branch capacity; pose versus shuffled isolates content. We call the latter the *content test* and treat it as decisive. Relative differences are $(A-B)/B$; negative favours A.

Inference is over *seeds*, which is the unit the treatment is applied to: every p-value and interval we report is a paired t-test on per-seed differences. We also report a hierarchical bootstrap that resamples seeds and then pedestrians, and it agrees; we do not quote a percentile bootstrap p-value, because with five seeds whose differences share a sign it returns the resolution floor $1/N_{\text{boot}}$ rather than a calibrated tail probability, and a pedestrian-only bootstrap ignores between-seed variance entirely.

C. Protocol

Every arm trains for 30 epochs with AdamW (batch 2 per GPU, weight decay 0.01, gradient-norm clipping 1.0) under a step schedule that halves the learning rate at epochs 10, 20 and 25; the decoder predicts $K{=}6$ modes. The learning rate is chosen per rung from a grid run separately for each arm: $1{\times}10^{-4}$ at 579 scenes, $5{\times}10^{-5}$ at 5k and $2.5{\times}10^{-5}$ at 25k, where every arm peaks at the same interior optimum. Validation is fixed at $5{,}171$ scenes and $16{,}486$ pedestrians; the directed search of Sec. IV-C splits it once into a development half ($2{,}617$ scenes, $8{,}445$ agents) and a held-out half ($2{,}554$ scenes, $8{,}041$ agents). Headline numbers are taken at each run's best validation epoch over a fixed evaluation schedule, and Sec. IV-B reports what changes under a fixed-epoch rule. Official WOMD mAP and gated miss rate need the full retained prediction set, which our per-epoch evaluation does not store, so we report minADE, minFDE and miss rate at 2 m. Code, configs, per-seed metrics and the analysis scripts are released with the paper.

IV. Results

A. The null, and its bound

Table I trains all three arms at three scales. Data alone improves the baseline by 19.2%; the content test is null at every rung, in minADE and in minFDE alike. At 25k the paired seed-level interval on the content difference is $[-1.59\%, +1.70\%]$,

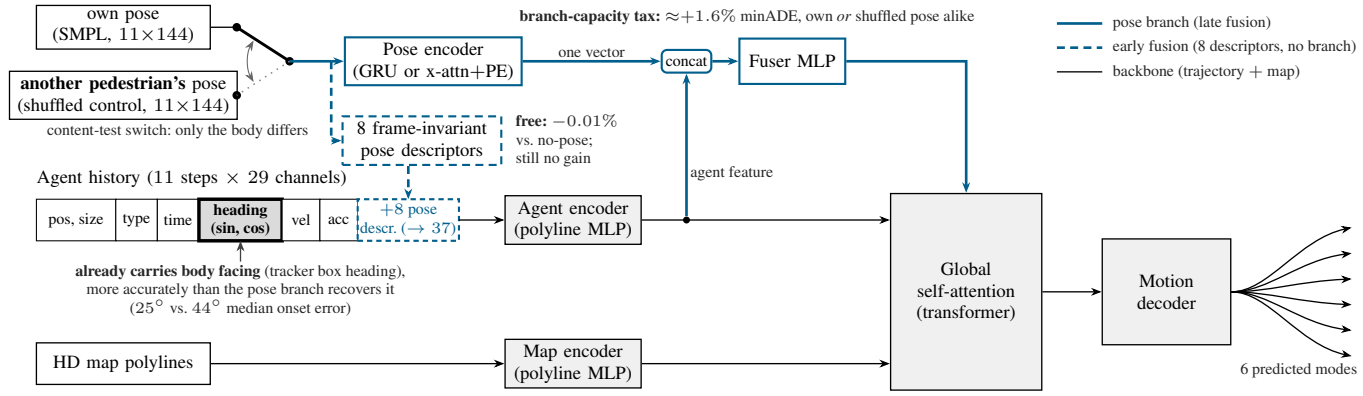


Fig. 1. The two arms differ only in *whose* body enters the model. The content-test switch (left) feeds the predictor either the pedestrian's own SMPL sequence or another pedestrian's, holding parameters, optimiser and input dimensionality fixed; everything downstream is shared. The agent-history strip shows that the tracker's box heading is *already* an input. Solid accent: the pose branch (late fusion), which costs ≈1.6% minADE whether it carries the right body or the wrong one; dashed: early fusion, which removes that cost and still yields no benefit.

TABLE I

CONTENT TEST ACROSS TRAINING SCALE, VALIDATION FIXED. THE LEARNING RATE IS SELECTED PER RUNG BY A GRID RUN SEPARATELY FOR EVERY ARM; ALL ARMS PEAK AT THE SAME VALUE (SEC. III-C). PERCENTAGES ARE DIFFERENCES OF ARM MEANS; p-VALUES ARE PAIRED t-TESTS OVER PER-SEED DIFFERENCES. NEGATIVE FAVOURS POSE.

Training scenes	579	5k	25k
Pedestrians	6,348	31,302	142,807
Seeds	5	3	5
minADE (m)			
No-pose	0.4599	0.4176	0.3716
Pose	0.4620	0.4138	0.3778
Shuffled	0.4589	0.4181	0.3777
Pose vs. no-pose	+0.45%	−0.90%	+1.66%*
Content test	+0.66%	−1.02%	+0.04%
p (content)	0.52	0.56	0.93
minFDE (m)			
No-pose	0.9922	0.8992	0.8049
Pose	0.9824	0.8875	0.8131
Shuffled	0.9780	0.8900	0.8104
Pose vs. no-pose	−0.99%	−1.30%	+1.02%†
Content test	+0.45%	−0.28%	+0.33%
p (content)	0.63	0.83	0.61

*$t(4)$=10.3, p=5×10^{-4}; the shuffled arm pays the same cost, +1.63%, $t(4)$=2.8, p=0.048.
†$t(4)$=6.1, p=0.004.

TABLE II

CLEAN CORE AT 579 SCENES, n=5 SEEDS, BOTH REGIMES. THE SHUFFLED ARM HAS THE LOWEST MEAN IN BOTH: ANOTHER PEDESTRIAN'S BODY IS NOMINALLY THE BEST INPUT.

Arm	Per-seed minADE	Mean ± sd	Content
From scratch			
No-pose	.458 .443 .453 .495 .450	0.4599 ± 0.0206	—
Shuffled	.454 .467 .451 .455 .467	0.4589 ± 0.0075	—
Pose	.471 .456 .453 .457 .474	0.4620 ± 0.0094	+0.66%
Fine-tuned on a full-WOMD backbone			
No-pose	.372 .366 .367 .373 .368	0.3692 ± 0.0031	—
Shuffled	.364 .363 .364 .374 .373	0.3676 ± 0.0055	—
Pose	.364 .372 .371 .374 .373	0.3708 ± 0.0040	+0.88%

so at the largest scale we train the data exclude a content effect beyond roughly 1.7% in either direction—a bound tighter than the 1.6% cost of the branch that would carry it. The pedestrian-resampling bootstrap gives a visibly narrower interval ($[-0.0040, +0.0047]$ m, $\approx \pm 1.2\%$); with five clusters it undercovers, and we quote the wider seed-level bound throughout.

After pretraining a trajectory-only backbone on full WOMD and fine-tuning (n=5), the content test is +0.88% (p=0.11), again with its sign against pose (Table II).

Seed count decides what Table II appears to say. Using only the first three seeds, the same runs read as *pose harms* from scratch (+1.85%) and *shuffled is best* after fine-tuning (−1.28%); at n=5 both shrink roughly fourfold, to +0.44% and −0.44%. The worst single run in the table—seed 404's no-pose run at 0.495—lies outside the three-seed set, and its absence flattered the baseline. The choice of inference unit compounds the budget: both three-seed effects read as significant under a pedestrian-resampling bootstrap (p=0.037 and p=0.016), while a paired t over the same three seeds gives p=0.18 and p=0.12. Three seeds do not resolve a 1% effect at this scale, and a bootstrap over the wrong unit will not tell you so.

B. Two effects that are real, and are not pose

A branch-capacity tax. At 25k both pose arms are worse than no-pose (+1.66%, p=5×10^{-4}; +1.63%, p=0.048) and indistinguishable from each other. An extra input branch costs ∼1.6% whether it carries the right body or the wrong one.

This is an operating-point statement, and the operating point matters: it is measured at each run's best validation epoch. Models with wider inputs overfit sooner, so under a fixed late-epoch rule the ordering reverses (+8.0% in favour of pose at epoch 30, p=0.30, because the no-pose arm has degraded further by then). We therefore never compare architecturally different arms at a fixed late epoch. The *content* test—whose two arms have identical input dimensionality, and for which this asymmetry does not arise—is null under either

rule. Replacing the 144-d branch with eight frame-invariant descriptors fused into the existing agent channels removes the cost entirely (-0.01% versus no-pose, p=0.999; -1.66% versus the branch, p=0.006) without creating a benefit: pose can be carried for free, and still does not pay.

Tuning dominates the treatment. At 5k the learning rate inherited from the smaller rung made pose look catastrophic ($+37\%$); with per-arm tuning, -0.90%, n.s. Across the 25k grid the learning rate moves minADE by up to 16.7% against a content signal of $\sim 0.05\%$: a pose-versus-baseline comparison that does not tune each arm separately is a claim about the learning rate.

C. Search and confirmation

Hand-designed integrations sample a small space, so we ran a directed search in which a coding agent rewrote the pose-integration code against a frozen training repository, scoring on a development split and never seeing the held-out split. Round one scored 13 candidates and accepted none. Round two produced one acceptance: a stationary-gated body-facing plus foot-contact descriptor that beat the incumbent on $5/5$ seeds and reached -1.21% content on the held-out split, clearing the pre-set 0.8% acceptance floor—under a truncated 12-epoch proxy recipe.

Because that acceptance rested on a truncated recipe, we pre-registered a confirmation before running it: three arms, eight seeds, 30 epochs, primary endpoint the held-out split at the fixed last epoch. It returned -0.04%, $t(7) = -0.11$, p=0.92. The proxy signal was a convergence-rate artifact—the arms differ in how fast they train, not in where they end—and a truncated proxy cannot settle a sub-1% effect. We report the acceptance rather than only the confirmation because the search behaved honestly: it found the best candidate the representation admits, and pre-registration is what removed it.

D. The branch is not inert

A null measured through one pathway invites the obvious objection that the pathway is dead. The shuffled control does not dispose of it: it shows the model cannot distinguish its own body from a stranger's, which is equally consistent with a branch that transmits nothing at all. The branch-capacity tax shows the branch perturbs the output, not that it carries signal, and the fusion residual is zero-initialised, so a dead branch and an uninformative one begin at the same place.

We therefore ran the positive control. The *oracle* arm replaces the SMPL features with a six-channel encoding of the agent's own future displacement at 3 and 8 s — genuinely predictive by construction — keeping the branch, its parameter count, its input dimensionality and its per-frame validity masks exactly as the pose arm has them, so the oracle is visible only where a real pose would have been.

Over three seeds the oracle arm reaches 0.2945 against no-pose's 0.4514 on the same seeds—-34.8%, $t(2) = -81.4$, $p = 1.5\times10^{-4}$—and -35.6% against the shuffled arm. The oracle is of course not a usable model, since it reads the future; its only job is to calibrate the pathway. It does so emphatically: the branch converts information into accuracy at roughly twenty times the magnitude of the bound our content test places on pose. The null is therefore a statement about what SMPL pose contains, not about whether the architecture can use what it is given.

E. Where a benefit could still hide

On the confirmation runs the content test is null at every horizon and least favourable at the shortest ($+0.48\%$ at 2 s, -0.04% at 8 s); null in all 20 tests across quartiles and top deciles of constant-velocity residual, heading variation, speed range and peak acceleration; and null for standing-then-moving pedestrians, the subpopulation built to favour it (n=1,809, $+0.38\%$, p=0.80, epoch-30 checkpoints on an expanded validation split).

V. Why Pose Does Not Pay

A. A detection floor for auxiliary channels

The content test bounds what pose is worth, and the oracle of Sec. IV-D shows the branch can carry information. Neither says how *much* information a side channel must carry before this pathway turns it into accuracy. We measured that directly.

Each rung of Table III feeds the branch the same direction-only oracle—the bearing of the agent's own future displacement at 3 and 8 s—corrupted by a fixed angular error, and is labelled by the median bearing error actually delivered. Displacement magnitude is withheld so the threshold can be stated in degrees. The error is drawn once per pedestrian rather than per epoch, because a real estimator's error is a property of the observation; resampling it each epoch would let training average it away and flatter a noisy channel.

The improvement decays cleanly, halving for every 4.4° of added bearing error, and is gone by 14.6°. Fitting the rungs that carry signal,

$$\Delta minADE(\theta) \;=\; -16.0\% \cdot \exp(-\theta/6.41^\circ). \qquad (1)$$

Past 14.6° the branch is not merely useless but harmful ($+2.7\%$ at 45°): the informational gain has decayed while the branch-capacity tax of Sec. IV-B has not. Setting the two equal puts break-even at 13.3°, which is where the measured curve changes sign.

B. Pose is far below the floor

How accurate is body facing in the units of (1)? We measured it on the task where pose should be at its best: which way a stationary pedestrian steps off, the one case where velocity carries no directional information. Over every pedestrian stationary at prediction time (< 0.3 m/s)—224,946 in training, 19,939 in validation, of whom 8.5% move more than 1 m within 3 s—SMPL facing predicts onset direction at roughly twice chance: median error 44.2°, 50.6% within 45° (chance 90°, 25%).

Genuinely informative, and still far below the floor. Equation (1) at 44.2° predicts an improvement of 0.02% against a capacity cost of $\approx 2\%$. Pose sits about eight halvings below

TABLE III

DETECTION FLOOR FOR THE POSE BRANCH. A DIRECTION-ONLY ORACLE CORRUPTED BY A FIXED PER-AGENT ANGULAR ERROR, EACH RUNG LABELLED BY THE MEDIAN BEARING ERROR ACTUALLY DELIVERED; n=3 SEEDS, PAIRED t AGAINST NO-POSE, NEGATIVE FAVOURS THE ORACLE. THE IMPROVEMENT HALVES EVERY 4.4° AND THE BRANCH TURNS INTO A NET COST PAST $\approx 13^\circ$. SMPL BODY FACING, AT 44.2°, LIES IN THE PENALTY REGION.

Median bearing error	ΔminADE	p	95% CI
0.0°	-15.7%	0.021	$[-25.6, -5.9]$
5.5°	-7.6%	0.0004	$[-8.2, -6.9]$
10.2°	-3.7%	0.084	$[-8.7, +1.2]$
14.6°	$+0.2\%$	0.876	$[-5.2, +5.6]$
27.2°	-0.1%	0.954	$[-5.4, +5.2]$
37.9°	$+0.6\%$	0.624	$[-3.8, +5.0]$
45.0°	$+2.7\%$	0.252	$[-4.6, +10.0]$
57.4°	$+2.7\%$	0.237	$[-4.2, +9.5]$

TABLE IV

PER-JOINT RECOVERABILITY AGAINST GROUND-TRUTH KEYPOINTS. MP IS THE ERROR OF ALWAYS PREDICTING THE MEAN POSE (EQUAL TO THE JOINT'S GROUND-TRUTH SPREAD BY CONSTRUCTION); $\mathrm{SKILL} = 1 - \mathrm{MPJPE}/\mathrm{MP}$. THE HEAD'S MPJPE BEATS BOTH WRISTS WHILE CARRYING A TENTH OF THEIR SKILL. INTERVALS RESAMPLE PEDESTRIANS (n=534), NOT FRAMES, SINCE FRAMES WITHIN A TRACK ARE STRONGLY CORRELATED. EVERY LIMB INTERVAL CLEARS ZERO BY A WIDE MARGIN (LOWEST BOUND $+0.43$); THE HEAD'S IS $[-0.003, +0.089]$, THE ONE INTERVAL THAT CONTAINS IT.

Joint	MPJPE (mm)	MP (mm)	SKILL
Elbows	91.2 / 91.5	252.9 / 248.9	+0.64 / +0.63
Hips	39.8	97.9	+0.59
Wrists	118.3 / 125.4	271.1 / 276.4	+0.56 / +0.55
Shoulders	87.3 / 89.0	171.5 / 164.0	+0.49 / +0.46
Ankles	114.0 / 116.6	221.6 / 223.8	+0.49 / +0.48
Knees	78.4 / 81.5	150.8 / 151.2	+0.48 / +0.46
Head	**111.6**	**116.9**	**+0.045**

the useful range, inside the region where the ladder measures a net penalty. The null is not a near miss.

Two measurements corroborate that 44.2° is the real figure rather than an artifact of our axis fitting. The tracker's own box heading estimates the same quantity at 25.2° and 63.9% within 45°; paired on the same events SMPL is 7.3° worse in median (Wilcoxon p=2.6×10^{-43}) and better on only 35% of events, even though we calibrated the SMPL forward axis *against* the box heading first. And pose does not take over where perception degrades: over $1{,}705$ onset events the gap favours the box heading in all ten testable strata ($p < 10^{-5}$ each) and *widens* with range, from $+12.3^\circ$ at 2–17 m to $+32.9^\circ$ at 41–75 m, because both estimates derive from the same LiDAR returns. Even in the jitteriest-heading quartile, where the box degrades to its worst value of 50.5°, SMPL is worse still at 63.8°.

What this does not establish. The box heading is *also* below the floor—(1) at 25.2° predicts 0.31%—so it is not the case that pose loses to heading because heading is more accurate. Through a late-fused side branch neither accuracy would pay. The box heading earns its place by being a native per-timestep input the model trains with from the start, a different delivery mechanism, and we have not run a ladder on that channel; the floor we report is a property of the auxiliary branch, not of every way a cue can enter a model.

We tested the implied counterfactual directly. Removing the heading channel entirely and repeating the content test changes it by -0.6 pp in the stationary subpopulation (n=8, p=0.65, CI $[-3.5, +2.3]$): no effect, which is what (1) predicts for a channel that was never above the floor. At n=3 that same comparison read -4.1 pp with all three seeds agreeing in sign—the third multi-percent effect in this study to dissolve under more seeds.

C. The complementary cue is barely measured

Gaze precedes locomotion, so head yaw relative to the torso is a plausible independent cue. It is very nearly not recoverable here. In the fitted meshes head-versus-torso yaw has a standard deviation of 3.8°; against the ground-truth 3D keypoints released with Waymo Perception [15], [16] (n=8,234) the real quantity varies 19.2° — a fivefold under-dispersion, so the estimator is largely emitting its prior. Consistently, head yaw carries no usable signal for onset direction ($r = -0.03$, n.s.), and the head joint's recoverability is the one value in Table IV whose interval includes zero. Head *position* is weakly recovered; head *orientation*, the quantity a gaze cue would need, is not. Physically, a pedestrian at 20–40 m returns a few dozen points, and the head is a near-spherical handful of them with little geometry to constrain yaw.

D. MPJPE is the wrong acceptance metric

A single mean error cannot distinguish an estimator that tracks a joint imprecisely from one that emits its prior. We therefore score each joint against the strongest trivial competitor, always predicting the mean root-aligned pose: $\mathrm{SKILL}_j = 1 - \mathrm{MPJPE}_j/\mathrm{MP}_j$, where values at or below zero mean the estimator contributes nothing at that joint.

Evaluated on Waymo Perception against ground-truth keypoints (1,789 frames, 534 pedestrians), every limb joint scores $+0.457$ to $+0.639$ (Table IV). The head scores $+0.045$—yet its MPJPE (111.6 mm) is *better* than either wrist (118.3/125.4 mm). Its error is flattering not because it is well estimated but because it barely improves on its own mean-pose baseline (116.9 mm), whereas the hips—whose ground-truth spread is *smaller* at 97.9 mm—are predicted at 39.8 mm. MPJPE rank-orders these joints roughly backwards with respect to information content. We recommend reporting per-joint skill alongside it whenever a pose estimator is accepted into a downstream stack.

E. A general condition

Two results locate where pose could have helped. At 2 s, minADE is 0.091 m against roughly 2.8 m of travel—about 3% of the displacement—so almost no error is left for an extra input to remove. We state that as a property of the forecaster's own residual, not of constant velocity: measured on the same validation set, single-mode constant-velocity extrapolation [17] gives an ADE of 0.310 m at 2 s and 1.459 m at 8 s, so the model is already $3.4\times$ better than constant

velocity at the short horizon and short-horizon forecasting is *not* CV-saturated. Saturation here means the baseline's remaining error is a few percent of the displacement, which is the condition that matters for an auxiliary modality. At long horizons the residual is dominated by destination and intent, which body configuration does not disclose. Stratifying instead by how badly constant-velocity extrapolation fails—the sharpest definition of "not saturated"—still yields no effect.

We therefore state the condition in general form: *an auxiliary modality improves a forecast only where the baseline is neither saturated nor dominated by intent the modality cannot observe, and where the modality's own measurement error leaves it above the integration pathway's detection floor.* All three clauses must hold. The third is the cheapest to check and the one most often skipped: measure the modality's accuracy on the cue it is supposed to carry, and compare it against a floor established with an oracle ladder—no modality needs to be integrated to find out that it cannot pay. For pedestrian forecasting on WOMD the window is empty on the third clause by a factor of four.

VI. What This Means in Practice

For an AV stack, three consequences follow. First, before integrating an auxiliary modality, measure its error on the cue it is meant to carry and compare that against the pathway's floor—an oracle ladder costs a handful of runs and needs no new sensor. Second, the capacity cost of an extra branch is real ($\sim$1.6% here, at the best-epoch operating point) and is easily mistaken for a modality effect in either direction; early fusion avoids it, and a shuffled-pose control separates the two. Third, do not accept a pose estimator on aggregate MPJPE alone: check per-joint skill, because the joints an aggregate flatters may be the ones carrying a prior.

Limitations. The largest rung we train is 25k scenes; the corrected export covers 299,220, so the full rung is a compute limit rather than an annotation one. At eight seeds the realised seed-to-seed spread gives a minimum detectable content effect of $\approx$1.25% at 80% power. We test one forecaster, one dataset and one pose estimator, and the floor we measure is a property of a late-fused side branch, not of every way a cue can enter a model. WOMD ships camera data only as quantised tokens, so a camera-based pose or gaze estimate cannot be tested at this scale; on Waymo Perception, where raw imagery exists, an RGB estimator (HMR2.0) reaches 106.2 mm root-aligned MPJPE against LiDAR-HMR's 87.6 mm on the same 347 frames, so the obvious better-pose route is the worse one here. We do not model pose–pose interaction.

VII. Conclusion

Body pose, recovered from LiDAR, does not improve pedestrian trajectory forecasting on WOMD. The null is controlled against a shuffled-pose baseline and holds across scale, representation, integration, a directed search and a pre-registered confirmation; an oracle through the same branch rules out a dead pathway. It also has a quantitative mechanism: this integration path pays only for bearing accurate to $\approx 15^\circ$, and body facing is four times coarser than that. The lesson generalises past pose—measure a modality's error on the cue it is meant to carry, and compare it against the pathway's floor, before integrating it.

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